

TP : LANGUAGE de REQUETE Relationnel // MongoDB :
--

⇒ *On travaille sur une collection nommée people*

```
{
  _id: ObjectId("509a8fb2f3f4948bd2f983a0"),
  user_id: "abc123",
  age: 55,
  status: 'A'
}
```

TRADUIRE LA REQUETE SQL en REQUETE MongoDB

-- 1

SELECT user_id, age, status

FROM people

WHERE age >= 40

ORDER BY status DESC;

db.people.find({ age: { \$gte: 40 } }, { user_id: 1, age: 1, status: 1 }).sort({ status: -1 });

-- 2

SELECT _id, user_id, age

FROM people

WHERE status IN ('A', 'B')

ORDER BY age ASC

LIMIT 5;

db.people.find({ status: { \$in: ["A", "B"] } }, { _id: 1, user_id: 1, age: 1 }).sort({ age: 1 }).limit(5);

-- 3

SELECT DISTINCT user_id

FROM people

MongoDB

WHERE status = 'A'

AND age BETWEEN 30 AND 60;

db.people.distinct("user_id", { status: "A", age: { \$gte: 30, \$lte: 60 } });

-- 4

SELECT _id, user_id, age

FROM people

WHERE status = 'A'

AND age > 50

ORDER BY age DESC;

db.people.find({ status: "A", age: { \$gt: 50 } }, { _id: 1, user_id: 1, age: 1 }).sort({ age: -1 });

-- 5

SELECT user_id, status, age

FROM people

WHERE status = 'A'

AND user_id LIKE 'a%'

AND age < 60;

db.people.find({ status: "A", user_id: /^a/, age: { \$lt: 60 } }, { user_id: 1, status: 1, age: 1 });

-- 6

SELECT user_id, age, status

FROM people

WHERE status NOT IN ('A')

AND age >= 18

ORDER BY user_id;

MongoDB

```
db.people.find({ status: { $nin: ["A"] }, age: { $gte: 18 } }, { user_id: 1, age: 1, status: 1
}).sort({ user_id: 1 });
```

-- 7

```
SELECT user_id, age
```

```
FROM people
```

```
WHERE status = 'A'
```

```
AND age BETWEEN 50 AND 70
```

```
ORDER BY age ASC
```

```
LIMIT 10;
```

```
db.people.find({ status: "A", age: { $gte: 50, $lte: 70 } }, { user_id: 1, age: 1 }).sort({ age:
1 }).limit(10);
```

-- 8

```
SELECT user_id, status, age
```

```
FROM people
```

```
WHERE status = 'A'
```

```
OR (age >= 45 AND age <= 60)
```

```
ORDER BY status, age;
```

```
db.people.find({ $or: [ { status: "A" }, { age: { $gte: 45, $lte: 60 } } ] }, { user_id: 1, status:
1, age: 1 }).sort({ status: 1, age: 1 });
```

-- 9

```
SELECT *
```

```
FROM people
```

```
WHERE age > 25
```

```
AND status IN ('A', 'C')
```

```
ORDER BY age DESC;
```

```
db.people.find({ age: { $gt: 25 }, status: { $in: ["A", "C"] } }).sort({ age: -1 });
```

-- 10

SELECT user_id, age

FROM people

WHERE age < 25

AND user_id LIKE '%3';

db.people.find({ age: { \$lt: 25 }, user_id: /3\$/ }, { user_id: 1, age: 1 });

-- 11

SELECT _id, user_id, age

FROM people

WHERE age BETWEEN 26 AND 50

AND status <> 'X'

ORDER BY user_id DESC;

db.people.find({ age: { \$gte: 26, \$lte: 50 }, status: { \$ne: "X" } }, { _id: 1, user_id: 1, age: 1 }).sort({ user_id: -1 });

-- 12

SELECT user_id, age

FROM people

WHERE user_id LIKE 'a%'

AND age >= 30;

db.people.find({ user_id: /^a/, age: { \$gte: 30 } }, { user_id: 1, age: 1 });

-- 13

SELECT user_id, age, status

FROM people

WHERE status = 'A'

MongoDB

AND age > 40

ORDER BY age ASC, user_id ASC;

```
db.people.find({ status: "A", age: { $gt: 40 } }, { user_id: 1, age: 1, status: 1 }).sort({ age: 1, user_id: 1 });
```

-- 14

SELECT user_id, age

FROM people

WHERE status = 'A'

AND age <= 60

ORDER BY age DESC, _id DESC;

```
db.people.find({ status: "A", age: { $lte: 60 } }, { user_id: 1, age: 1 }).sort({ age: -1, _id: -1 });
```

-- 15

SELECT COUNT(*) AS total_active

FROM people

WHERE status = 'A';

```
db.people.countDocuments({ status: "A" });
```

-- 16

SELECT COUNT(DISTINCT user_id) AS unique_users

FROM people

WHERE age > 30;

```
db.people.distinct("user_id", { age: { $gt: 30 } }).length;
```

-- 17

SELECT status, COUNT(*) AS count_by_status

MongoDB

FROM people

WHERE age >= 30

GROUP BY status;

```
db.people.aggregate([
  { $match: { age: { $gte: 30 } } },
  { $group: { _id: "$status", count: { $sum: 1 } } }
]);
```

-- 18

SELECT DISTINCT status, age

FROM people

WHERE age > 40;

```
db.people.find({ age: { $gt: 40 } }, { status: 1, age: 1, _id: 0 });
```